A WEIGHTING CONDITION ON A TWISTED OBJECT CAN BE CHECKED AFTER AN ETALE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 22110-22122 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 22124-22149, the complete author proof (24 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Part of the chapter's weighted-pictures series; the proof is a self-contained degree computation with no deferred steps.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definitions of the weighting data of the chapter's picture/grading section (stabilizer group actions, degrees on tangent space pieces) and elementary facts that degrees of polynomials are unchanged by field extension.

Source reference. Original lemma 09HJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22149–22149. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Given a cartesian square*

$$\begin{array}{ccc} U & \xleftarrow{h} & U' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ V & \xleftarrow{g} & V' \end{array}$$

with π locally quasi-finite with finite fibres and a function $w : U \to \mathbf{Z}$ we have $(\int_\pi w) \circ g = \int_{\pi'} (w \circ h)$.

Proof. This follows immediately from the second description of $\int_\pi w$ given above. To prove it from the definition, you use that if E/F is a finite extension of fields and F'/F is another field extension, then writing $(E \otimes_F F')_{red} = \prod E'_i$ as a product of fields finite over F', we have

$$[E : F]_s = \sum [E'_i : F']_s$$

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To prove this equality pick an algebraically closed field extension Ω/F' and observe that

$$\begin{aligned}
[E : F]_s &= |\operatorname{Mor}_F(E, \Omega)| \\
&= |\operatorname{Mor}_{F'}(E \otimes_F F', \Omega)| \\
&= |\operatorname{Mor}_{F'}((E \otimes_F F')_{red}, \Omega)| \\
&= \sum |\operatorname{Mor}_{F'}(E'_i, \Omega)| \\
&= \sum [E'_i : F']_s
\end{aligned}$$

where we have used Fields, Lemma 09HJ. □